

9.4 Volatility

A lower volatility in FMP indicates a greater certainty in estimating the cost of quality control, whether through peer reviews, replications, or otherwise. A healthy system is expected to develop, eventually, better methods that help in estimating these costs, and hence a trend towards lower volatility is expected.

Consider $\bar{s}_p(t)$ be the global peer review fair market price. Then, volatility in peer reviewer marketplace over a time period $\{t - (n - 1), \dots, t - 1, t\}$ can be computed as

$$\text{Volatility}_{p,n} = \sigma_{\bar{s}_{p,n}} = \sqrt{\mathbb{E}[(\bar{s}_p(t) - \mathbb{E}[\bar{s}_p(t)])^2] \cdot n} \quad (77)$$

Similarly, the volatility for the replicators marketplace can be computed as

$$\text{Volatility}_{r,n} = \sigma_{\bar{s}_{r,n}} = \sqrt{\mathbb{E}[(\bar{s}_r(t) - \mathbb{E}[\bar{s}_r(t)])^2] \cdot n} \quad (78)$$

9.5 Global research efficiency

This is the efficiency of research considering all the manuscripts in the world, i.e., replace Π with Θ in eq. (48), eq. (49), i.e., the amount of academic capital produced by the globe as a whole normalized by the total amount of research spending (or the total time of). At the global scale, the normalization on funding could also be done by other global economic measures, e.g., GDP, PPP, etc.

$$\text{Funding Efficiency}_{\Theta} = \epsilon_{\$, \Theta} = \frac{\mathbb{X}_{\Theta}}{\$_{\Theta}} \quad (79)$$

$$\text{GDP Efficiency}_{\Theta} = \epsilon_{\text{GDP}, \Theta} = \frac{\mathbb{X}_{\Theta}}{\$_{\Theta}} \cdot \text{GDP}_{\Theta} \quad (80)$$

$$\text{PPP Efficiency}_{\Theta} = \epsilon_{\text{PPP}, \Theta} = \frac{\mathbb{X}_{\Theta}}{\$_{\Theta}} \cdot \text{PPP}_{\Theta} \quad (81)$$

$$\text{Time Efficiency}_{\Theta} = \epsilon_{t, \Theta} = \frac{\mathbb{X}_{\Theta}}{\Delta t_{\Theta}} \quad (82)$$

This metric tries to explain whether the increase in global academic capital is due to increase in research spending, or due to the methodology of research becoming efficient as well. The global research system is expected to streamline its processes, enabled by better quantification of capital output and improvised quality control mechanisms. Hence, the research efficiency of the globe is expected to increase overall.

All the global research efficiency metrics described above can also be computed for smaller geographic regions $\theta_x \in \Theta$. This allows to investigate the research efficiency in a given geographical area, over a given period of time, so that the influence of policy and governance on academic output could also be measured.

9.6 Transaction volume

This is a measure of the total number of quality control actions (peer review or replications) that happens in a unit time. This gives an estimate of the amount of quality control happening in the system. Consider $\Pi_{\Delta t, \Theta}$ be the portfolio of all manuscripts produced in Δt time period over the globe. Then, the transaction volume would be computed as the total number of peer reviewer and replicator nodes on those manuscripts. This can be easily computed by fetching $\hat{M} \in \Pi$ subset of manuscripts from the mask tensor $\mathcal{M}$'s peer reviewer and replicator slices, and taking a sum over the values.

9.7 Time efficiency of quality control

The amount of time peer review and replication take will be different across different academic fields, but should not have variability based on geographical location. Let $t_{p,d}(m)$ denote the time taken for peer review of manuscript m in some academic field d , and $t_{r,d}(m)$ denote the time taken for replication of the manuscript. These values represent the sum total of all peer reviews and replications involved, if there were multiple peer reviewers or replicators. Then, the time efficiency for quality control in a field d is computed by taking a mean of the times over the subset $\hat{M}_D$ of manuscripts in D ,

$$\epsilon_{p,D} = \frac{1}{|\hat{M}_D|} \sum_{\hat{M}_D} t_{p,d}(m) \quad (83)$$

$$\epsilon_{r,D} = \frac{1}{|\hat{M}_D|} \sum_{\hat{M}_D} t_{r,d}(m) \quad (84)$$

In a healthy system, we anticipate that the time efficiency of quality control will gradually increase, as better quality control is proportionately rewarded.

9.8 Collection Subscription Ratio

One way of measuring whether the academic impact of papers is only within it's own field, or outside of the academic field, is to look at the ratio of the number of collection subscribers to the number of authors in that collection. Recall from section 6.13 that a collection is defined by a set of tags $\hat{D}$ that could be from the 4D academic field tags or from filters like peer reviewed, replicated, particular authors, institutions, geographic regions, or time periods. Manuscripts with tags D_m such that $\hat{D} \subseteq D_m$ automatically appear in that collection. The CSR for that collection is defined as:

$$\text{CSR}(\hat{D}) = \frac{\text{subscribers to } \hat{D}}{\text{authors in } \hat{D}} \quad (85)$$

For an academic field, the higher this value, the more impactful the work in this academic field is said to be outside of that academic field. For a journal, the higher this value, the more influential that journal's work is said to be beyond the journal's own contributor community.

10 Exploits & Modifications

Any choice of how to count academic capital, there may be exploits or inadequacies with the $\mathbf{X}$ metric. Research impactfulness is a multidimensional quantity with different components and weightings for different people, much like how value generated in an economy also has many definitions beyond the raw currency value. Likewise with econometrics, overly focusing on any single one allows policy making to become biased in its optimization goals, potentially to detrimental effects. In practical circumstances, a suite of econometrics are used to portray a more complete picture of reality, and the same practice is encouraged for Liberata despite having a default, flagship way of calculating $\mathbf{X}$. Liberata’s academic capital calculations are designed to be modular, with information about how $\mathbf{X}$ is counted stored in the references graph (section 4) and information about how $\mathbf{X}$ is to be split and attributed amongst contributors stored in the shares graph (section 3). This allows the swapping of either graph with newly defined graphs without affecting other graphs and metrics. To construct a suite of $\mathbf{X}$ variants to paint a more complete picture of academic impact, several proposed variants of the references graph $\mathbf{G}_w$ are given below, each designed to cover potential weaknesses or oversight regions of the others.

10.1 Time Modulated Weighted Citations

The first candidate modification replaces the weighted citations references graph matrix $\mathbf{G}_w$ with a time modulated weighted citations (TMWC) references graph matrix $\mathbf{G}_{wT}$ according to the operation below. This results in a version of capital assignment that allows larger duration (longer project time) works to give out more proportionately more capital than shorter duration works. This is an important distinction than other prior works Walker et al. [2007], Wang et al. [2013], Parolo [2015] that take temporal component into consideration for citation weighting, as the proposed modification is more adaptive to different publication cycles of different disciplines and allows arbitrary time-dependent transformations.

$$\mathbf{G}_{wT} = \mathbf{G}_w \circ \begin{bmatrix} \frac{t_{3,1}}{t_{3,2}} \\ \frac{t_{3,2}}{\vdots} \\ \frac{t_{3,|M|}}{\vdots} \end{bmatrix} \quad (86)$$

Here, the average time to publication t_3 at the discipline tag d_3 level is averaged from prior literature of the same d_3 primary tag, and each row (corresponding to a manuscript) of $\mathbf{G}_w$ is scaled by multiplying the original value by the row manuscript’s corresponding t_3 to yield $\mathbf{G}_{wT}$. While slightly harder to interpret from a given number, this version of capital assignment is a reasonable alternative to the simple weighted citations capital assignment method. For $\mathbf{X}$ to be counted in this way, a counter-incentive is needed to guard against artificial inflation of project time for more academic credit, and we propose the time efficiency in section 6.10 metric for this. In addition, this modification, along with others, are not meant to be used in isolation and are instead meant to be used in unison to depict a more accurate picture of academic impact.

10.2 Impact Modulated Weighted Citations

A second candidate modification, called impact modulated weighted citations (IMWC) scales up the citation power of each article according to their impact, measured using the default definition of $\mathbf{X}$ in eq. (10). To ensure asymptotic convergence and to prevent distortions from long self citation chains, rows of the weighted citations references graph $\mathbf{G}_w$ are scaled by an impact modifier ι with a logarithmically decaying construction. This construction is related to recursive prestige-based citation weighting methods such as PageRank and Pinski–Narin influence models Page et al. [1999], Pinski and Narin [1976], Chen et al. [2007], but differs in explicitly modulating the citation adjacency matrix using a nonlinear, log-damped impact transformation, and in employing a truncated iterative scheme.

$$\iota(m \in M) = \log_2(\mathbf{X}_m + 1) \quad (87)$$

$$\mathbf{G}_{w\iota} = \mathbf{G}_w \circ \begin{bmatrix} \frac{\iota_1}{\iota_2} \\ \frac{\iota_2}{\vdots} \\ \frac{\iota_{|M|}}{\vdots} \end{bmatrix} \quad (88)$$

This alternative to assigning academic capital scales up the citation power of works logarithmically according to their own impactfulness, rewarding works that lead to impactful works more. Importantly, the causal

chain is kept short by the logarithmic nature of this impact factor definition, preventing large distortions to second degree ancestral works and beyond. This definition also implies an iterative solver, but the iterations can be truncated after four steps due to the fourth logarithm being so insignificantly small that we practically reach the asymptotic value after three or four steps.

10.3 Author Cosine Similarity Discounting

A third candidate modification is to discount $\star$ going from authors to themselves, also known as self citation. In this method, we construct a matrix of author cosine similarity modifiers (ACSM):

$$\Phi = \begin{bmatrix} 0 & \phi_{1,2} & \cdots & \phi_{1,n-1} & \phi_{1,n} \\ 0 & 0 & \cdots & \phi_{2,n-1} & \phi_{2,n} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & \phi_{n-1,n} \\ 0 & 0 & \cdots & 0 & 0 \end{bmatrix} \quad (89)$$

with each element defined as:

$$\phi_{k,l}(\vec{s}'_k, \vec{s}'_l) = \begin{cases} \vec{s}'_k \cdot \vec{s}'_l & \text{if } m_l \text{ cites } m_k \\ 0 & \text{otherwise} \end{cases} \quad (90)$$

Here, k and l are two indices corresponding to two manuscripts m_k and m_l , and $\vec{s}'_k$ and $\vec{s}'_l$ are the extended shares split vectors for m_k and m_l respectively, constructed via the process below.

1. Take the union of authors in m_k and m_l , $A_{k,l} = A_k \cup A_l$
2. Assign each $a \in A_{k,l}$ a unique index n .
3. Set the length of $\vec{s}'_k$ and $\vec{s}'_l$ to be of size n .
4. Assign index n of $\vec{s}'_k$ and $\vec{s}'_l$ to be the shares that author a_n holds in m_k and m_l respectively.

The weighted citations references graph $\mathbf{G}_W$ is then multiplied element wise by $\mathbb{1} - \Phi$ to yield the ACSM references graph, where $\mathbb{1}$ is the ones matrix:

$$\mathbf{G}_{W\Phi} = \mathbf{G}_W \circ (\mathbb{1} - \Phi) \quad (91)$$

This modification attempts to discourage self citing by discounting citations coming from works by the similarity of the author list share distributions. This modification generalizes the idea of fractional self-citation proposed by Schubert et al. [2006] by using author-share vectors to compute a continuous edge-level discount, so that citations are penalized in proportion to the similarity of contribution distributions across the citing and cited manuscripts. This modification, like the others, is not meant to be used in isolation, but instead is meant to give independent information in conjunction with other modifications.

10.4 Alternative Modifications

The modular nature of Liberata's references graph also allows for the linear modifications above, which are commutative and associative, to be chained together in any order to form compound modifications.

The modifications above are expected to be a small subset of possible and sensible modifications to how $\star$ is counted. As other modifications are developed, they too can be swapped in or combined with the ones defined above. Such modifications might try to take into account information about intellectual property generated from the research, or attempt to quantify downstream economic impacts, or even propitiatory custom constructs similar to Altmetrics. Priem et al. [2012] It is hoped that a suite of future variants created by the academic community will illuminate the landscape of academic contributions if/when the Liberata system sees increasing adoption.